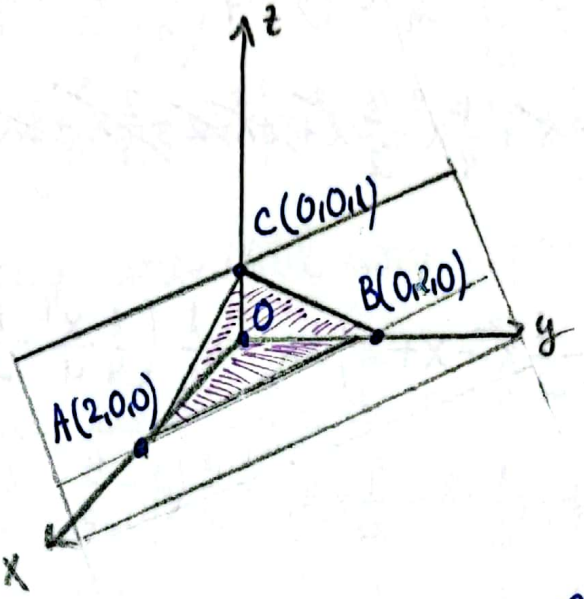
Integrale triple

• Să se calculeze următoarele integrale triple

$\iiint_\Omega f(x,y,z)\,dxdydz$:

(1) Ω este mulțimea mărginită de planele $x=0, y=0, z=0$ și planul $\alpha: x+y+2z=2$; $f(x,y,z)=x+y+z$.



Determinăm punctele de intersecție ale planului α cu axele de coordonate: $A(2,0,0)$; $B(0,2,0)$; $C(0,0,1)$.

Avem $0 \le z \le \frac{1}{2}(2-x-y)$

$$\Rightarrow \iiint_\Omega f(x,y,z)dxdydz = \iint_D \left(\int_0^{\frac{1}{2}(2-x-y)} (x+y+z)dz \right) dxdy =$$

$$= \iint_D \left(xz+yz+\frac{z^2}{2} \Big|_0^{\frac{1}{2}(2-x-y)} \right) dxdy =$$

$$= \iint_D \left(\frac{1}{2}(2x-x^2-xy)+\frac{1}{2}(2y-xy-y^2)+\frac{1}{8}(2-x-y)^2 \right) dxdy =$$

$$= \frac{1}{2} \iint_D \left(2x+2y-x^2-y^2-2xy+1+\frac{1}{4}x^2+\frac{1}{4}y^2-x-y+\frac{1}{2}xy\right)dxdy =$$

$$= \frac{1}{2} \iint_D \left(\frac{-3}{4}(x^2+y^2)-\frac{3}{2}xy+x+y+1\right)dxdy \overset{not}{=} I$$

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Avem $D = pr_{xOy}\Omega = \Delta OAB$, unde $O(0,0)$, $A(2,0)$, $B(0,2)$

$\Rightarrow D = \{(x,y) \in \mathbb{R}^2 \mid x \in [0,2];\ 0 \le y \le 2-x\}$

$$\Rightarrow I = \frac{1}{2}\int_0^2 \left(\int_0^{2-x} \left(-\frac{3}{4}(x^2+y^2) - \frac{3}{2}xy + x + y + 1 \right) dy \right) dx =$$

$$= \frac{1}{2}\int_0^2 \left[\left(-\frac{3}{4}x^2y - \frac{3}{4}\cdot\frac{y^3}{3} - \frac{3}{4}xy^2 + xy + \frac{y^2}{2} + y \right)\Big|_0^{2-x} \right] dx =$$

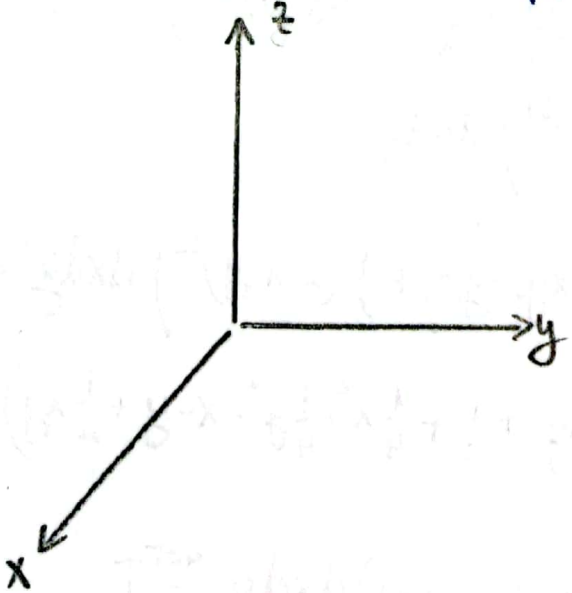
$$= \frac{1}{2}\int_0^2 \left[-\frac{3}{4}x^2(2-x) - \frac{1}{4}(2-x)^3 - \frac{3}{4}x(2-x)^2 + x(2-x) + \frac{1}{2}(2-x)^2 + 2 - x \right] dx =$$

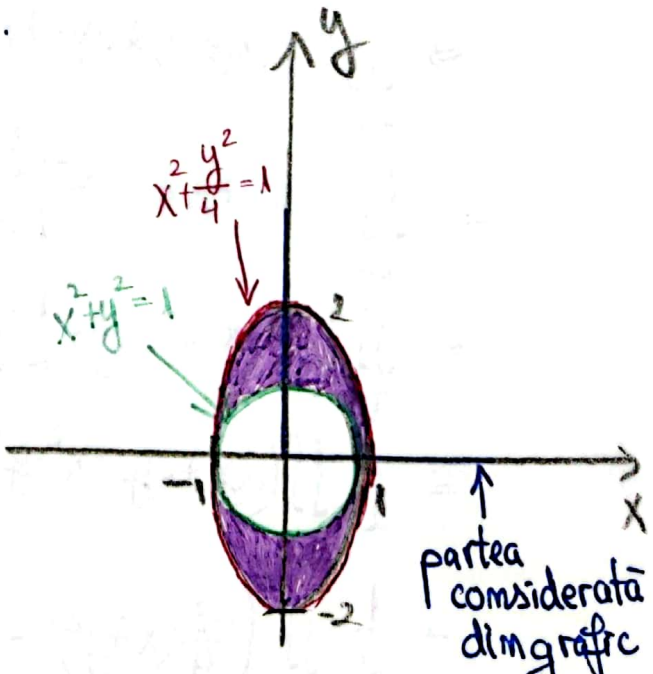
$$= \frac{1}{2}\int_0^2 \Big(\cancel{\frac{3}{4}x^3} - \cancel{\frac{3}{2}x^2} + \frac{1}{4}x^3 - \cancel{\frac{3}{2}x^2} + \cancel{3x} - \cancel{2} - \cancel{\frac{3}{4}x^3} + \cancel{3x^2} - \cancel{3x} + \cancel{2x} - x^2 + \frac{x^2}{2} - \cancel{2x} + \cancel{2} + \cancel{2} - x \Big) dx =$$

$$= \frac{1}{2}\int_0^2 \left(\frac{1}{4}x^3 - \frac{1}{2}x^2 - x + 2 \right) dx = \frac{1}{2}\left(\frac{1}{4}\cdot\frac{x^4}{4} - \frac{1}{2}\cdot\frac{x^3}{3} - \frac{x^2}{2} + 2x \right)\Big|_0^2 =$$

$$= \frac{1}{2}\left(\frac{1}{16}\cdot 16 - \frac{1}{6}\cdot 8 - \frac{4}{2} + 4 \right) = \frac{1}{2}\left(1 - \frac{4}{3} - 2 + 4\right) = \frac{5}{6}.$$

(2) $\Omega = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + \frac{y^2}{4} \le 1;\ x^2+y^2 \ge 1;\ x \ge 0;\ y \ge 0;$
$0 \le z \le 5\}$; $f(x,y,z) = \frac{yz}{\sqrt{x^2+y^2}}$.





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$x^2+\frac{y^2}{4}\le 1$ este interiorul unui cilindru eliptic, de semiaxe $a=1$ și $b=2$, cu axa de simetrie Oz.

$x^2+y^2\ge 1$ este exteriorul unui cilindru drept, cu secțiunea cerc de rază $R=1$.

$$\iiint_\Omega f(x,y,z)\,dxdydz = \iint_D \left(\int_0^5 \frac{yz}{\sqrt{x^2+y^2}}\,dz\right)dxdy =$$

$$= \iint_D \left(\frac{y}{\sqrt{x^2+y^2}}\cdot\frac{z^2}{2}\Big|_0^5\right)dxdy = \frac{25}{2}\iint_D \frac{y}{\sqrt{x^2+y^2}}\,dxdy \overset{not}{=} I$$

$$D = pr_{xOy}\,\Omega = \left\{(x,y)\in\mathbb{R}^2 \,\middle|\, x^2+\frac{y^2}{4}\le 1,\ x^2+y^2\ge 1;\ x\ge 0;\ y\ge 0\right\}$$

$x^2+\frac{y^2}{4}\le 1$ este interiorul unei elipse de semiaxe $a=1$ și $b=2$

$x^2+y^2\ge 1$ este ext. cercului de centru $C(0,0)$ și raza $R=1$.

$x^2+\frac{y^2}{4}\le 1 \Leftrightarrow y\le 2\sqrt{1-x^2}$; $x^2+y^2\ge 1 \Leftrightarrow y\ge\sqrt{1-x^2}$

$$\Rightarrow D = \left\{(x,y)\in\mathbb{R}^2 \,\middle|\, x\in[0,1];\ \sqrt{1-x^2}\le y\le 2\sqrt{1-x^2}\right\}.$$

$$\Rightarrow I = \frac{25}{2}\int_0^1\left(\int_{\sqrt{1-x^2}}^{2\sqrt{1-x^2}}\frac{y}{\sqrt{x^2+y^2}}\,dy\right)dx. = \frac{25}{2}\int_0^1\left(\int_{\sqrt{1-x^2}}^{2\sqrt{1-x^2}}\left(\sqrt{x^2+y^2}\right)'dy\right)dx$$

$$= \frac{25}{2}\int_0^1\left(\sqrt{x^2+y^2}\Big|_{\sqrt{1-x^2}}^{2\sqrt{1-x^2}}\right)dx = \frac{25}{2}\int_0^1\left(\sqrt{x^2+4-4x^2}-\sqrt{x^2+1-x^2}\right)dx$$

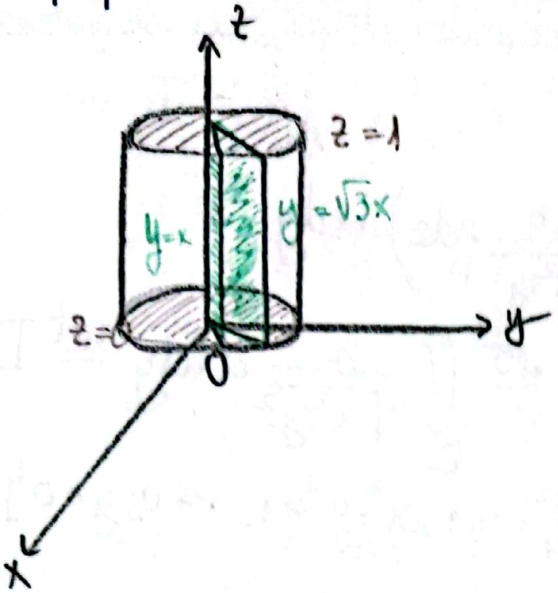
$$= \frac{25}{2}\int_0^1\sqrt{4-3x^2}\,dx - \frac{25}{2}.$$

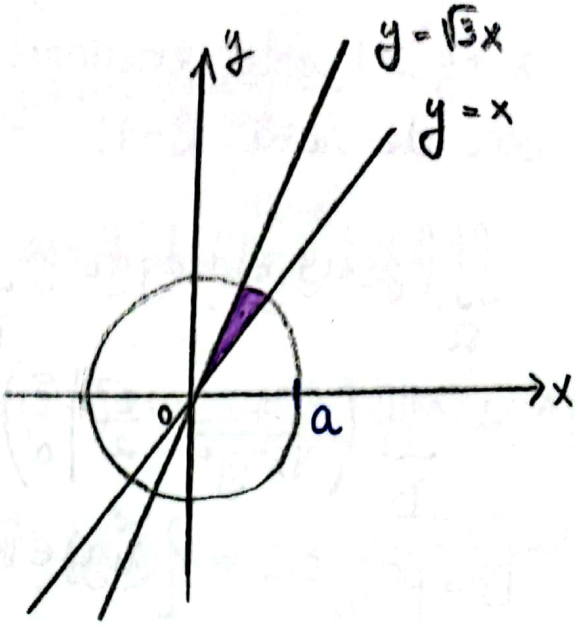
$$\int_0^1\sqrt{4-3x^2}\,dx = \int_0^1\frac{4-3x^2}{\sqrt{4-3x^2}}\,dx = 4\int_0^1\frac{1}{\sqrt{4-3x^2}}\,dx - 3\int_0^1\frac{x^2}{\sqrt{4-3x^2}}\,dx$$

(...) $I = \frac{25}{12\sqrt{3}}\left(4\pi - 3\sqrt{3}\right)$.

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③ Ω este mulțimea mărginită de cilindrul $x^2+y^2=a^2$ și planele $z=0$, $z=1$, $y=x$, $y=\sqrt{3}x$; $f(x,y,z)=xy$





$x^2+y^2=a^2$ este ecuația unui cilindru eliptic cu axa de simetrie Oz.

$$I=\iiint_{\Omega} f(x,y,z)\,dxdydz=\iint_{D}\left(\int_0^1 xy\,dz\right)dxdy=$$

$$=\iint_{D} xy\,dxdy$$

$$D=pr_{xOy}\Omega=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le a^2;\ x\le y\le\sqrt{3}x\}$$

$x^2+y^2\le a^2$ este interiorul cercului de centru $(0,0)$ și raza $R=a$

Facem schimbarea de variabilă și trecem la coordonate polare:

$$\begin{cases} x=r\cos t \\ y=r\sin t \end{cases} \Rightarrow r\in[0,a];\ t\in\left[\tfrac{\pi}{4},\tfrac{\pi}{3}\right];\ J(r,t)=r$$

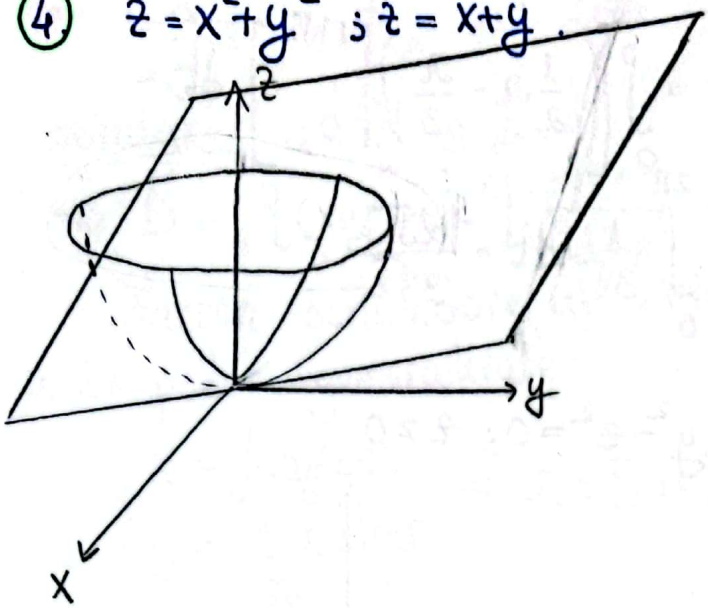
$$\Rightarrow I=\int_{\pi/4}^{\pi/3}\left(\int_0^a r^3\sin t\cos t\,dr\right)dt=\int_{\pi/4}^{\pi/3}\sin t\cos t\left[\left(\frac{r^4}{4}\right)\Big|_0^a\right]dt=$$

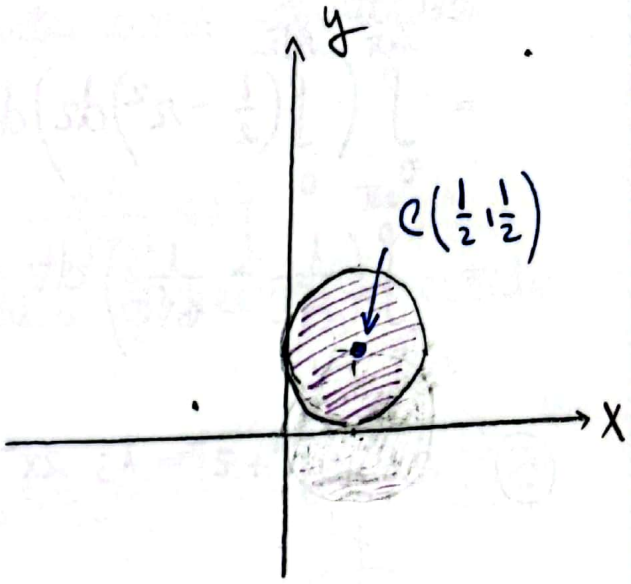
$$=\int_{\pi/4}^{\pi/3}\sin t\cos t\cdot\frac{a^4}{4}\,dt=\frac{a^4}{8}\int_{\pi/4}^{\pi/3}\sin 2t\,dt=\frac{a^4}{8}\cdot\frac{\cos 2t}{2}\Big|_{\pi/4}^{\pi/3}=\frac{a^4}{32}$$

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• Să se calculeze volumul mulțimilor Ω mărginite de suprafețele încadrate (cu reprezentare grafică):

④ $z = x^2+y^2$; $z = x+y$.





$x^2+y^2 = z$ este paraboloid eliptic cu vârful în origine și axa de simetrie Oz.

$x+y-z = 0$ este un plan care trece prin origine

$$V = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{x^2+y^2}^{x+y} dz \right) dx\,dy = \iint_D (x+y-x^2-y^2)\,dx\,dy$$

$D = pr_{xOy}\,\Omega$

$x^2+y^2 = x+y \Leftrightarrow x^2-x+\frac{1}{4}-\frac{1}{4}+y^2-y+\frac{1}{4}-\frac{1}{4} = 0 \Leftrightarrow$

$\Leftrightarrow \left(x-\frac{1}{2}\right)^2 + \left(y-\frac{1}{2}\right)^2 = \frac{1}{2}$ este ecuația unui cerc de centru $C\left(\frac{1}{2}, \frac{1}{2}\right)$ și raza $\frac{1}{\sqrt{2}}$.

$\Rightarrow D = \left\{(x,y) \in \mathbb{R}^2 \,\middle|\, \left(x-\frac{1}{2}\right)^2 + \left(y-\frac{1}{2}\right)^2 \leq \left(\frac{1}{\sqrt{2}}\right)^2\right\}$

Facem schimbarea de variabilă și trecem la coordonatele polare generalizate

$$\begin{cases} x = \frac{1}{2} + r\cos t \\ y = \frac{1}{2} + r\sin t \end{cases} \Rightarrow r \in \left[0, \frac{1}{\sqrt{2}}\right] ; t \in [0, 2\pi] ; \underline{J(r,t) = r}$$

$$\Rightarrow V = \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} (x+y-x^2-y^2)\,dr \right) dt = \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} \left(1 + r\sin t + r\cos t - \right.\right.$$

$$\left.\left. - \left(\frac{1}{2}+r\cos t\right)^2 - \left(\frac{1}{2}+r\sin t\right)^2 \right) dr \right) dt \overset{J.}{=}$$

! MAI TREBUIA ÎNMULȚIT CU r !

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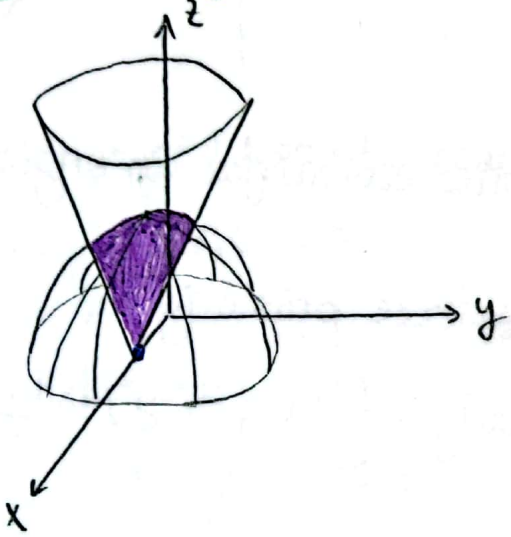
$$= \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} \left(1 + r\sin t + r\cos t - \frac{1}{4} - r\cos t - r^2\cos^2 t + \frac{1}{4} - r\sin t - r^2 \sin \right.\right.$$

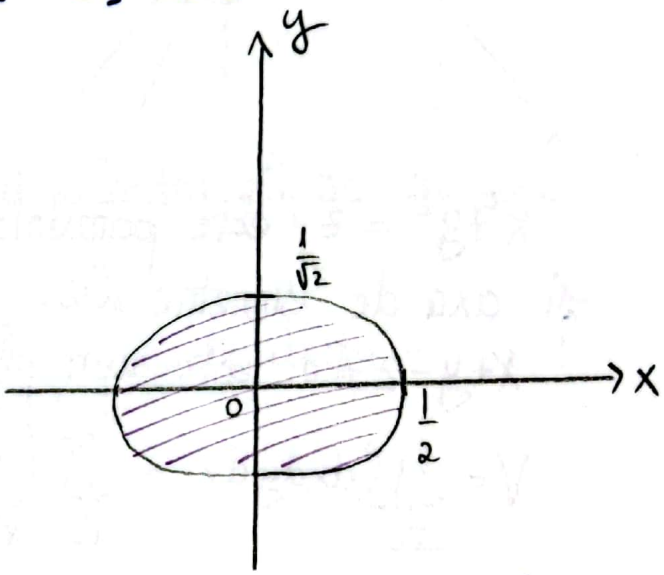
$$\left.\left. dr\right) dt. =$$

$$= \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} \left(\frac{1}{2} - r^2\right) dr \right) dt = \int_0^{2\pi} \left[\left(\frac{1}{2} r - \frac{r^3}{3} \right) \Big|_0^{1/\sqrt{2}} \right] dt =$$

$$= \int_0^{2\pi} \left(\frac{1}{2\sqrt{2}} - \frac{1}{6\sqrt{2}} \right) dt = \int_0^{2\pi} \frac{1}{3\sqrt{2}} dt = \frac{2\sqrt{2}\pi}{3\sqrt{2}}$$

(5) $2x^2 + y^2 + z^2 = 1$; $2x^2 + y^2 - z^2 = 0$; $z \geq 0$





$2x^2 + y^2 + z^2 = 1 \Leftrightarrow \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + y^2 + z^2 - 1 = 0$ ec unui elipsoid

$2x^2 + y^2 - z^2 = 0 \Leftrightarrow \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + y^2 - z^2 = 0$ ec unui con eliptic

$$V = \iiint_\Omega \underbrace{f(x,y,z)}_{1} dx\,dy\,dz = \iint_D \left(\int_{\sqrt{2x^2+y^2}}^{\sqrt{1-2x^2-y^2}} dz \right) dx\,dy =$$

$$= \iint_D \left(\sqrt{1-2x^2-y^2} - \sqrt{2x^2+y^2} \right) dx\,dy.$$

$D = pr_{xOy}\,\Omega$

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$2x^2+y^2-z^2=0 \Leftrightarrow z^2=2x^2+y^2$

$\Rightarrow 2x^2+y^2+z^2=1 \Leftrightarrow 2x^2+y^2+2x^2+y^2=1 \Leftrightarrow 4x^2+2y^2=1 \Leftrightarrow$

$\Leftrightarrow \frac{x^2}{\left(\frac{1}{2}\right)^2}+\frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2}=1$ este ecuația unui elipsoid de semiaxe $a=\frac{1}{2}$ și $b=\frac{1}{\sqrt{2}}$.

$\Rightarrow D=\left\{(x,y)\in\mathbb{R}^2 \mid \frac{x^2}{(1/2)^2}+\frac{y^2}{(1/\sqrt{2})^2}\le 1\right\}$

Facem schimbare de variabilă și trecem la coordonate polare generalizate:

$$\begin{cases} x=\frac{1}{2}r\cos t \\ y=\frac{1}{\sqrt{2}}r\sin t \end{cases} \Rightarrow r\in[0,1];\ t\in[0,2\pi];\ J(r,t)=\frac{r}{2\sqrt{2}}$$

$$\Rightarrow V=\int_0^{2\pi}\left(\int_0^1 \frac{r}{2\sqrt{2}}\left(\sqrt{1-2\cdot\frac{1}{4}r^2\cos^2 t-\frac{1}{2}\cdot r^2\sin^2 t}-\sqrt{2\cdot\frac{1}{4}r^2\cos^2 t+\frac{1}{2}r^2\sin^2 t}\right)dr\right)dt=$$

$$=\int_0^{2\pi}\left(\int_0^1 \frac{r}{2\sqrt{2}}\left(\sqrt{1-\frac{1}{2}r^2}-\sqrt{\frac{1}{2}r^2}\right)dr\right)dt=$$

$$=\int_0^{2\pi}\left(\int_0^1\left[\frac{r}{2\sqrt{2}}\sqrt{1-\frac{1}{2}r^2}-\frac{r^2}{4}\right]dr\right)dt=$$

$$=\int_0^{2\pi}\left(\int_0^1 \frac{r}{2\sqrt{2}}\sqrt{\frac{2-r^2}{2}}\,dr-\int_0^1\frac{r^2}{4}dr\right)dt.$$

$$\int_0^1 \frac{r^2}{4}dr=\frac{1}{4}\cdot\frac{r^3}{3}\Big|_0^1=\frac{1}{4}\cdot\frac{1}{3}=\frac{1}{12}$$

$$\int_0^1 \frac{r}{2\sqrt{2}}\sqrt{\frac{2-r^2}{2}}\,dr=\int_0^1\frac{r}{4}\sqrt{2-r^2}\,dr=\frac{1}{4}\int_0^1 r\sqrt{2-r^2}\,dr=$$

$$=\frac{1}{4}\left(\frac{r^2}{2}\sqrt{2-r^2}\Big|_0^1-\int_0^1 r\cdot\frac{-2r}{2\sqrt{2-r^2}}dr\right)=\frac{1}{4}\left(\frac{1}{2}-\int_0^1\frac{-r^2}{\sqrt{2-r^2}}dr\right)=$$

$$=\frac{1}{4}\left(\frac{1}{2}-\int_0^1\frac{2-r^2-2}{\sqrt{2-r^2}}dr\right)=\frac{1}{4}\left(\frac{1}{2}-\int_0^1\frac{2-r^2}{\sqrt{2-r^2}}dr+2\int_0^1\frac{1}{\sqrt{2-r^2}}dr\right)$$

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$$\int_0^1 \frac{1}{\sqrt{2-x^2}}dx = \int_0^1 \frac{1}{\sqrt{(\sqrt{2})^2-x^2}}dx = \arcsin\frac{x}{\sqrt{2}}\Big|_0^1 = \frac{\pi}{4}$$

$$\int_0^1 \frac{2-x^2}{\sqrt{2-x^2}}dx = \int_0^1 \sqrt{2-x^2}dx = x\sqrt{2-x^2}\Big|_0^1 - \int_0^1 x\cdot\frac{-2x}{2\sqrt{2-x^2}}dx =$$

$$= \int_0^1 \frac{x^2}{\sqrt{2-x^2}}dx + 1 = 1 - \underbrace{\int_0^1 \frac{-x^2}{\sqrt{2-x^2}}dx}_{\text{not } I}$$

Avem: $\frac{1}{4}\left(\frac{1}{2}-I\right) = \frac{1}{4}\left(\frac{1}{2}-1+I+2\cdot\frac{\pi}{4}\right) \Rightarrow$

$\Rightarrow \frac{1}{8}-\frac{1}{4}I = -\frac{1}{8}+\frac{1}{4}I+\frac{\pi}{8} \Rightarrow \frac{1}{2}I = \frac{1}{4}-\frac{\pi}{8} \Rightarrow$

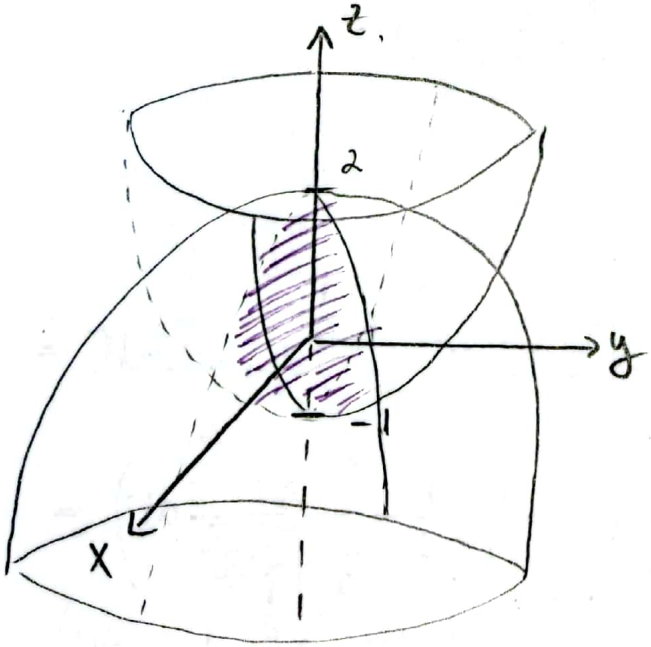
$\Rightarrow I = \frac{1}{2}-\frac{\pi}{4} = \frac{2-\pi}{4}$

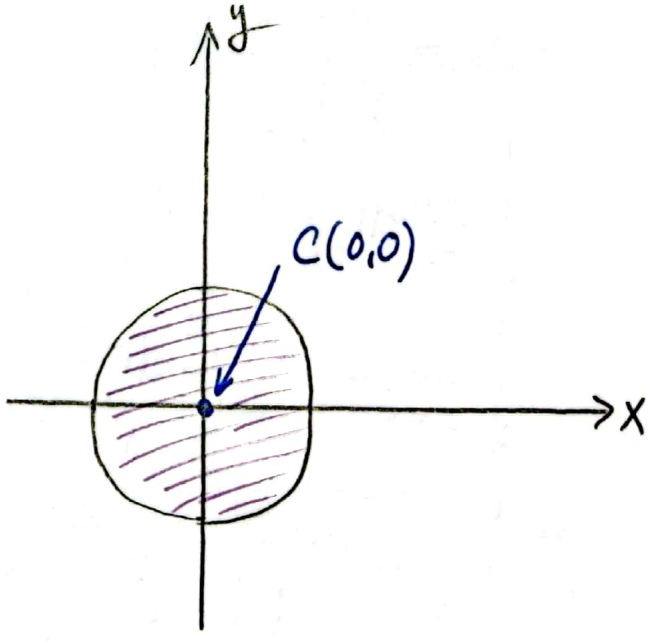
$$\Rightarrow \int_0^1 \frac{x}{2\sqrt{2}}\sqrt{\frac{2-x^2}{2}}dx = \frac{1}{4}\left(\frac{1}{2}-I\right) = \frac{1}{4}\left(\frac{1}{2}-\frac{1}{2}+\frac{\pi}{4}\right) = \frac{\pi}{16}$$

$$\Rightarrow V = \int_0^{2\pi}\left(\frac{\pi}{16}-\frac{1}{12}\right)dt = 2\pi\,\frac{4\pi-3}{48} = \frac{\pi(4\pi-3)}{24}$$

HABAR N-AM DACĂ E CORECT, DAR AM ÎNCERCAT :)

(6) $z = x^2+y^2-1$; $z = 2-x^2-y^2$





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$z = x^2+y^2-1 \Leftrightarrow x^2+y^2 = z+1$ este ecuația unui paraboloid eliptic cu vârful în $A(0,0,-1)$ și cu deschidere în sus.

$z = 2-x^2-y^2 \Leftrightarrow x^2+y^2 = \frac{z-2}{-1}$ este ecuația unui paraboloid eliptic cu vârful în $B(0,0,2)$ și cu deschidere în jos.

$$V = \iiint_\Omega dxdydz = \iint_D \left(\int_{x^2+y^2-1}^{2-x^2-y^2} dz \right) dxdy = \iint_D (2-x^2-y^2-x^2-y^2+1)dxdy =$$

$$= \iint_D (3-2x^2-2y^2)dxdy.$$

$z = x^2+y^2-1 = 2-x^2-y^2 \Leftrightarrow 2x^2+2y^2 = 3 \Leftrightarrow x^2+y^2 = \frac{3}{2}$ este ecuația unui cerc de centru $C(0,0)$ și raza $R = \sqrt{\frac{3}{2}}$.

$$\Rightarrow D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq \left(\sqrt{\tfrac{3}{2}}\right)^2\}$$

Facem schimbarea de variabilă și trecem la coordonate polare.

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r \in [0, \tfrac{\sqrt{6}}{2}]; \ t \in [0, 2\pi]; \ J(r,t) = r$$

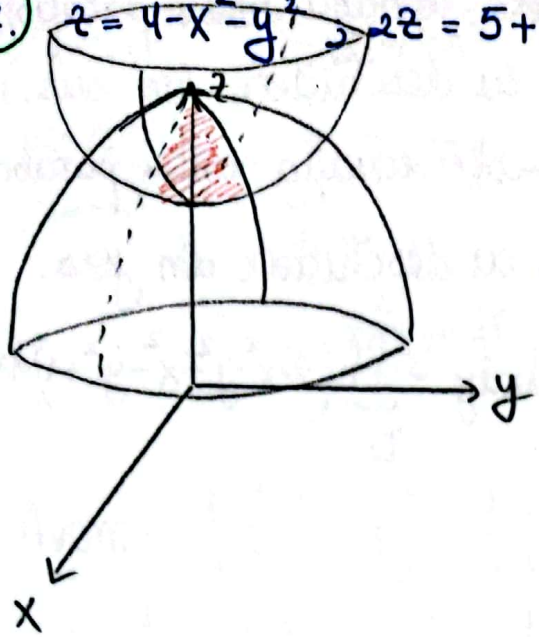
$$\Rightarrow V = \int_0^{2\pi} \left(\int_0^{\sqrt{6}/2} r \cdot (3 - 2r^2\cos^2 t - 2r^2\sin^2 t)\, dr \right) dt =$$

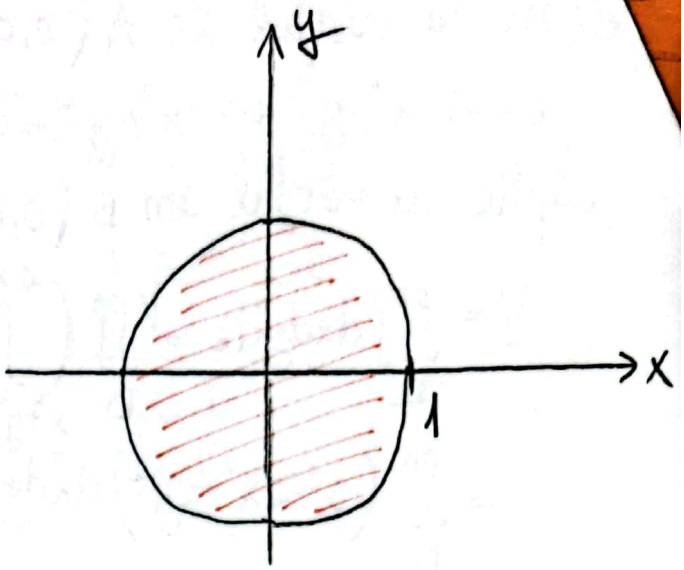
$$= \int_0^{2\pi} \left(\int_0^{\sqrt{6}/2} (3r - 2r^3)\, dr \right) dt = \int_0^{2\pi} \left[\left(\frac{3}{2}r^2 - \frac{2}{4}r^4 \right) \Big|_0^{\sqrt{6}/2} \right] dt =$$

$$= \int_0^{2\pi} \left(\frac{9}{4} - \frac{1}{2} \cdot \frac{36}{16} \right) dt = 2\pi \left(\frac{9}{4} - \frac{9}{8} \right) = 2\pi \cdot \frac{9}{8} = \frac{9\pi}{4}$$

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7. $z = 4 - x^2 - y^2$, $2z = 5 + x^2 + y^2$





$z = 4 - x^2 - y^2 \Leftrightarrow x^2 + y^2 = \frac{z-4}{-1}$: ecuația unui paraboloid eliptic cu vârful în $(0,0,4)$ și cu deschidere în jos

$2z = 5 + x^2 + y^2 \Leftrightarrow x^2 + y^2 = \frac{z - \frac{5}{2}}{\frac{1}{2}}$: ecuația unui paraboloid eliptic cu vârful în $(0,0,\frac{5}{2})$ și cu deschidere în sus.

$$V = \iiint_\Omega dxdydz = \iint_D \left(\int_{\frac{5+x^2+y^2}{2}}^{4-x^2-y^2} dz \right) dxdy = \iint_D \left(4 - x^2 - y^2 - \frac{5+x^2+y^2}{2} \right) dxdy =$$

$$= \iint_D \left(\frac{3}{2} - \frac{3}{2}x^2 - \frac{3}{2}y^2 \right) dxdy = \frac{3}{2} \iint_D (1 - x^2 - y^2)\, dxdy$$

$D = pr_{xOy}\, \Omega$

$z = 4 - x^2 - y^2 = \frac{5 + x^2 + y^2}{2} \Leftrightarrow 8 - 2x^2 - 2y^2 = 5 + x^2 + y^2 \Leftrightarrow$

$\Leftrightarrow 3x^2 + 3y^2 = 3 \Leftrightarrow x^2 + y^2 = 1$: ec unui cerc de centru $C(0,0)$ și raza $R = 1$

$D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$

Facem schimbarea de variabilă și trecem la coordonate polare:

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r \in [0,1];\ t \in [0, 2\pi];\ J(r,t) = r.$$

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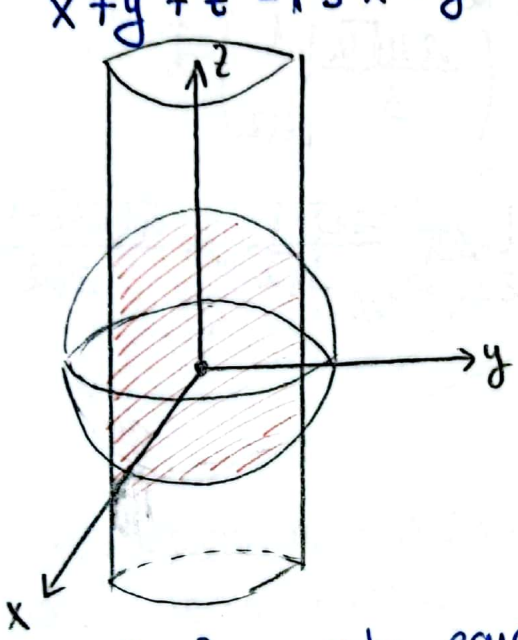
$$V=\frac{3}{2}\int_0^{2\pi}\left(\int_0^1 r\cdot\left(1-r^2\cos^2t-r^2\sin^2t\right)dr\right)dt =$$

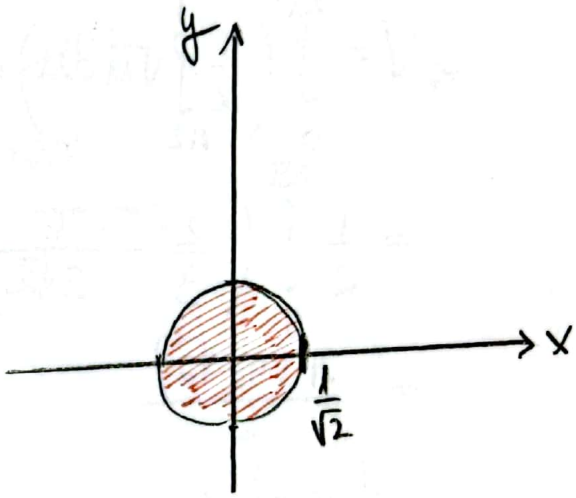
$$=\frac{3}{2}\int_0^{2\pi}\left(\int_0^1 r(1-r^2)dr\right)dt=\frac{3}{2}\int_0^{2\pi}\left(\int_0^1(r-r^3)dr\right)dt=$$

$$=\frac{3}{2}\int_0^{2\pi}\left[\left(\frac{r^2}{2}-\frac{r^4}{4}\right)\Big|_0^1\right]dt=\frac{3}{2}\int_0^{2\pi}\left(\frac{1}{2}-\frac{1}{4}\right)dt=\frac{3}{2}\int_0^{2\pi}\frac{1}{4}dt=$$

$$=\frac{3}{8}\cdot 2\pi=\frac{3\pi}{4}.$$

8. $x^2+y^2+z^2=1 \;;\; x^2+y^2=\frac{1}{2}$





$x^2+y^2+z^2=1$ este ecuația unei sfere centrată în origine și de rază 1.

$$x^2+y^2=\frac{1}{2}\Leftrightarrow 2x^2+2y^2=1\Leftrightarrow\frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2}+\frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2}=1$$

este ecuația unui cilindru eliptic, ce secționea un cerc de rază $R=\frac{1}{\sqrt{2}}$.

Considerăm: $0\le z\le\sqrt{1-x^2-y^2}$

$$\Rightarrow V=\iiint_\Omega dx\,dy\,dz=2\iint_D\sqrt{1-x^2-y^2}\,dx\,dy.$$

$$D=pr_{xOy}\Omega=\left\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le\left(\frac{1}{\sqrt{2}}\right)^2\right\}$$

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Facem schimbare de variabilă și trecem la coordonate polare:

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r \in \left[0, \frac{1}{\sqrt{2}}\right]; \; t \in [0, 2\pi]; \; J(r,t) = r$$

$$\Rightarrow V = \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} r \cdot \sqrt{1 - r^2\cos^2 t - r^2 \sin^2 t}\, dr \right) dt =$$

$$= \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} r\sqrt{1-r^2}\, dr \right) dt$$

$$1 - r^2 = u \Rightarrow -2r\,dr = du \Rightarrow r\,dr = -\frac{1}{2}du$$

$$r = 0 \Rightarrow u = 1; \quad r = \frac{1}{\sqrt{2}} \Rightarrow u = \frac{1}{2}$$

$$\Rightarrow V = \int_0^{2\pi} \left(\frac{1}{2} \int_1^{1/2} \sqrt{u}\, du \right) dt = \frac{1}{2} \int_0^{2\pi} \left(\frac{2u\sqrt{u}}{3} \Big|_1^{1/2} \right) dt =$$

$$= \frac{1}{2} \int_0^{2\pi} \left(\frac{2}{3} - \frac{1}{3\sqrt{2}} \right) dt = \frac{1}{2} \cdot 2\pi \, \frac{2\sqrt{2}-1}{3\sqrt{2}} = \frac{\pi(2\sqrt{2}-1)}{3\sqrt{2}} =$$

$$= \frac{\pi(4-\sqrt{2})}{6}$$

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